Traffic Sign Detection System

A Traffic Sign Detection System is an Advanced Driver Assistance System (ADAS) component that uses vehicle-mounted cameras, computer vision, and deep learning—primarily convolutional neural networks (CNNs) and YOLO—to detect, recognize, and interpret road signs (speed limits, warnings) in real-time. It improves driving safety

- **Methodology:** Uses deep learning models YOLO (V4-V8) for speed and high accuracy .
- **Process:**
    1. **Image Acquisition:** Cameras capture images, which may be preprocessed to remove noise .
    2. **Detection/Localization:** Identifies if a sign exists and locates it.
    3. **Recognition/Classification:** Determines the specific type of sign.

Datasets

- German Traffic Sign Detection Benchmark (GTSDB).
- **Real-time Processing:** Optimized algorithms, such as E-mobilenet or MASG-Net, allow for high-speed, real-time detection at over 200 frames per second (FPS).

Project Overview:

This project implements a traffic sign detection system using YOLOv8 with advanced data augmentation techniques. The model is trained on the GTSDB (German Traffic Sign Detection Benchmark) dataset, which contains 900 images with 43 different traffic sign classes.

Key Features:

- Real-time traffic sign detection
- 43 traffic sign classes
- Advanced data augmentation for better generalization
- mAP50: 74.1% on test set

1. **GTSDB Dataset**

Total Images: 900

Training Images: 600

Test Images: 300

Classes: 43 traffic sign categories

Total Annotations: 1,206 traffic signs

2. **Model Specifications**

Component	Details
Base Model	YOLOv8m
Total Parameters	25.8 million
Layers	170
GFLOPs	79.2
Input Size	640×640
Output	Bounding boxes + Class probabilities

Paper:

YOLO-BS: a traffic sign detection algorithm based on YOLOv8

mailto:https://www.nature.com/articles/s41598-025-88184-0

DataSet:

https://benchmark.ini.rub.de/gtsdb_dataset.html

Yolov8:

https://arxiv.org/abs/2304.00501